

공학수학 I (BS102a-02)

학기	2026년 1학기
담당교수	Hudson Thomas Bryan
이수구분	공통필수
학점	3.0
강의시간	화16:30-18:30(E7 - 223), 목16:30-18:30(E7 - 223)
강의실	E7 L12 (교수실)
소속	기초학부

교과개요

Differential equations represent a fundamental mathematical tool used by the whole of science to model physical phenomena: they express how a certain quantity of interest is related to its derivatives. Solutions to such problems are given by differentiable functions, which can be used to provide estimates and predict the evolution of the physical system one is studying. In this course students will learn how to solve several types of ordinary differential equations and also how to investigate and interpret the results.

교과목표

Students are expected to critically apply the different methods of solution of differential equations presented in the course and to be able to interpret and effectively investigate the solutions obtained, both numerically and qualitatively.

주차별 강의계획

1주차:Differential equations and mathematical modeling
2주차:Review of differentiation and integration
3주차:Euler's method, numerical solutions to differential equations
4주차:Linear ODEs
5주차:Nonhomogeneous ODEs
6주차:Existence and uniqueness of solutions to IVP
7주차:Variation of parameters
8주차:Midterm exam
9주차:Basic theory of systems of ODEs
10주차:Constant-Coefficients systems of ODEs
11주차:Taylor series
12주차:Series solution of ODEs
13주차:Legendre's and Bessel's equations
14주차:Laplace transform
15주차:Applications of Laplace transform (if time allows)
16주차:Final Exam

수업방법

The course will mainly consist of frontal lectures, with some in-class opportunities for the students to test their understanding with some exercises. Homework exercises will be provided to prepare the students for the biweekly quizzes.

학습윤리

Cheating and plagiarism will not be tolerated and, if discovered, will lead to failing the entire course. This also applies to students who knowingly allow their work to be copied.

교과정책(방침)

In order to pass the course students are expected to score at least 20% on both the midterm and the final. In particular, not attending either the midterm or the final will result in a failing grade. Exceptions to this general rule will be granted on a case-by-case basis at the discretion of the instructor.

Make-up exams will be granted only in exceptional cases, following the submission of appropriate documentation explaining the circumstances.

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준비물 및 기타

Lecture notes and the recordings of the lectures will be made available on LMS.